

16 QAM Vector Modulator Design

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1. Introduction

1.1 Design Specifications

The vector modulator presented in this project was designed to support 16-QAM (Quadrature Amplitude Modulation) operation at a carrier frequency of 5.4 GHz. The required minimum double-sideband bandwidth was 1.8 GHz, corresponding to a symbol rate of 900 MHz, with four bits per symbol. Two bits control the phase state using $0^\circ/180^\circ$ modulation, while the remaining two bits determine the amplitude level. So in other words, two bits control the phase and two control the attenuation. The design requirements specified a maximum insertion loss of 8.5 dB and a maximum VSWR of 1.4:1 across the operating bandwidth. The resulting constellation was required to maintain a phase accuracy within $\pm 3.5^\circ$ and an amplitude variation within ± 0.7 dB over the bandwidth. To meet these constraints, the entire design was implemented as a purely passive circuit, using Rogers 4003 substrate material for the transmission line.

1.2 Design Assumptions

Several assumptions were made to simplify the initial design process. The four digital control bits used to define each constellation point were represented using ± 3 V and ± 1 V DC signals, where a positive voltage corresponded to a logic '1' and a negative voltage represented a logic '0'. While the required voltage inputs to the attenuators were 3V in magnitude, the inputs to the IF ports of the mixers were determined to be 1V in magnitude, following industry standards. A key assumption was that the DMX0418L mixer used in the phase modulator has a phase error of $\pm 0.5^\circ$ from the device tolerances while having proper 180 degree modulation operation. This

comes from the diode pairs in the structure of the mixer acting in opposition to each other, which is the primary principle behind phase modulation.

2. Vector Modulator Architecture

The overall vector modulator architecture followed the recommended block-level structure, while allowing flexibility in the implementation of each individual stage.

Figure 1 shows a high-level block diagram of the complete system, and Figure 2 illustrates the ideal 16-QAM constellation used as a reference for the design.

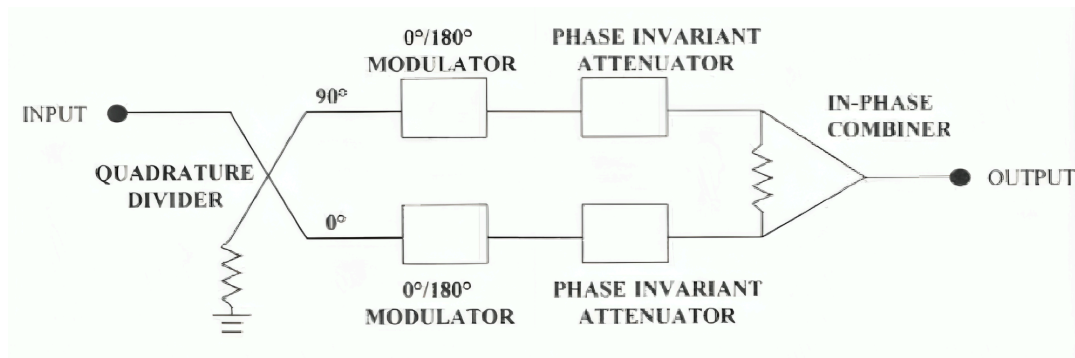


Figure 1. Recommended Block Diagram

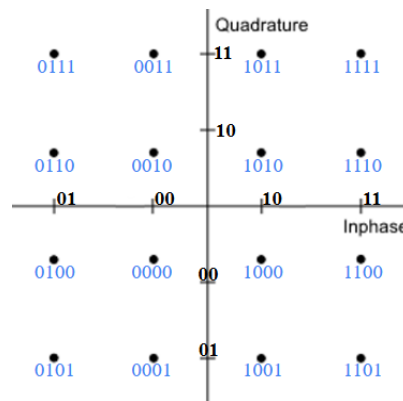


Figure 2. Ideal 16-QAM Constellation

For both the I and Q axes, the most significant bit (MSB) defines the signal polarity, and the least significant bit (LSB) determines the amplitude magnitude from the origin. So points closest to the origin are the most attenuated while the outer radius of the points are what is restricted by the minimum insertion loss of 8.5 dB.

2.1 Quadrature Divider

The first stage of the vector modulator is a quadrature divider, which splits the RF input signal into two equal-amplitude signals separated by 90° of phase. One output remains in phase with the input, while the second output is phase-shifted by 90° . Two quadrature divider topologies were evaluated, including branch line hybrid and Lange couplers. The branch line hybrid design is shown in Figures 3 and 4 provided 3.07 dB power division but exhibited VSWR variations across the 4.5-6.3 GHz sweep range, which complicated full system analysis in ADS. From an operational standpoint, this VSWR variation is not critical since the quadrature divider operates at the single frequency of 5.4 GHz, with a maximum bandwidth consideration of ± 20 MHz to account for non-idealities. However, during system-level simulations that swept the entire frequency range, the varying port impedances obscured the analysis of individual stage contributions and overall system behavior. Therefore, the Lange coupler was chosen as a better alternative that still met the frequency requirements while having a lower VSWR.

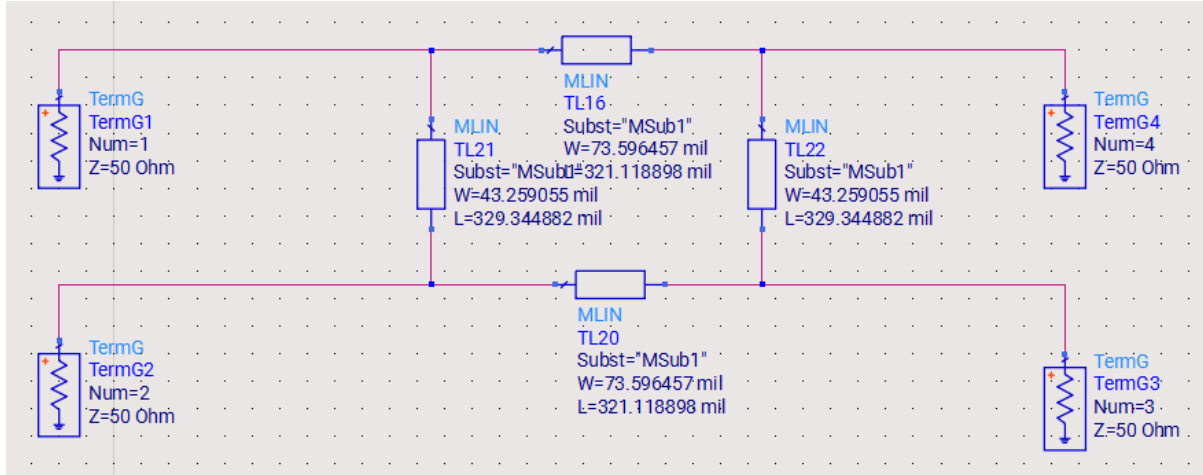


Figure 3. Branch Line Coupler Design on ADS

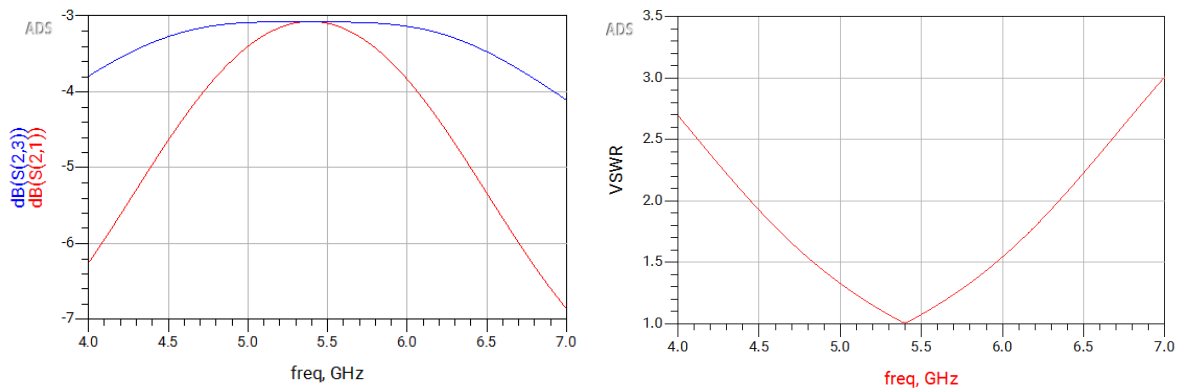


Figure 4. Branch Line Coupler Performance: Insertion Loss (Left) and VSWR (Right)

This topology uses multiple interleaved transmission-line fingers to achieve strong coupling with low insertion loss, 3.02 dB and 3.11 dB at the two output ports, low VSWR stability (0.01 variation across the simulation bandwidth), 90.15° phase shift at 5.4 GHz, high isolation (26.85 dB), and high return loss (26.6 dB). Detailed specifications are provided in Table 1 and Appendix A.

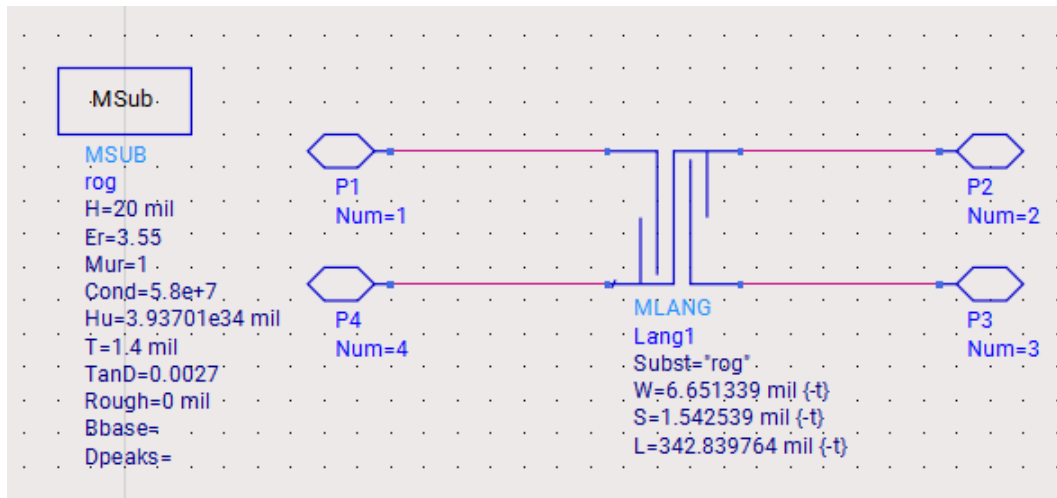


Figure 5. Lange Coupler Design on ADS

Lange Coupler Output Plots

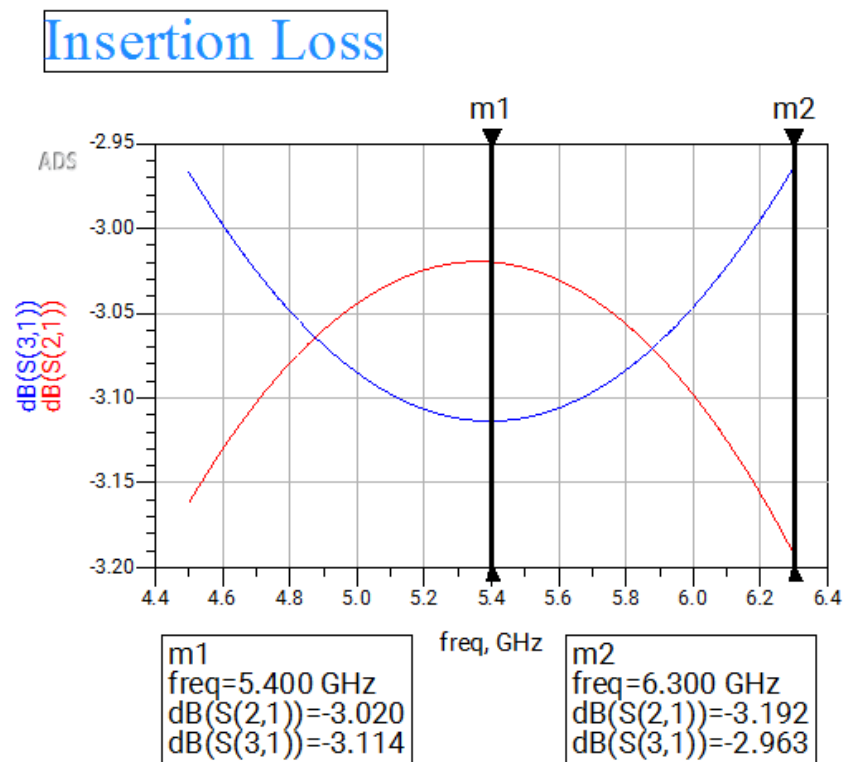


Figure 6. Lange Coupler Insertion Loss

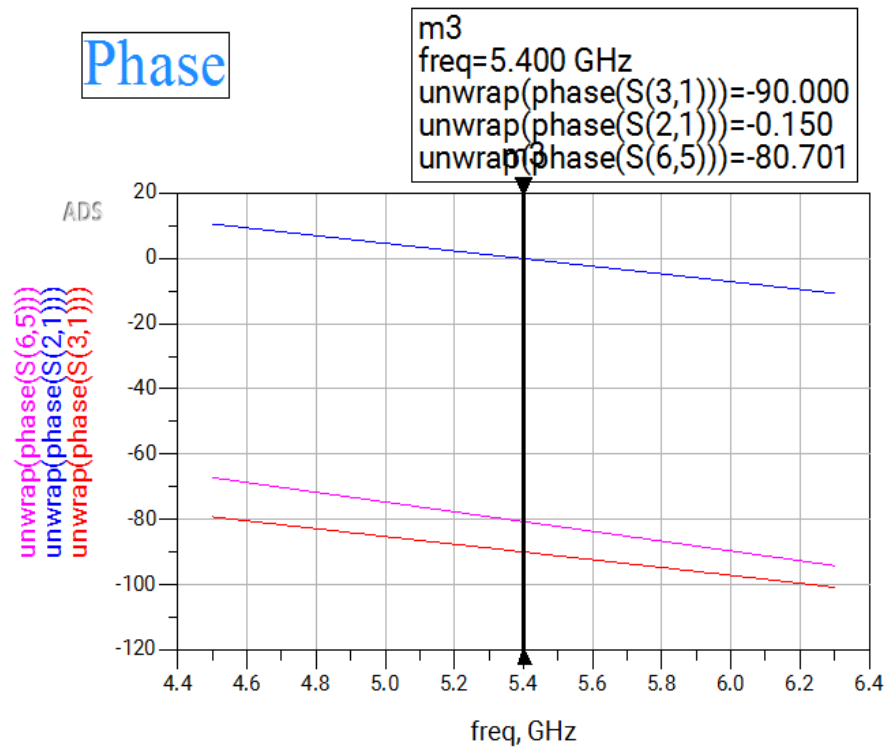


Figure 7. Lange Coupler Insertion Loss

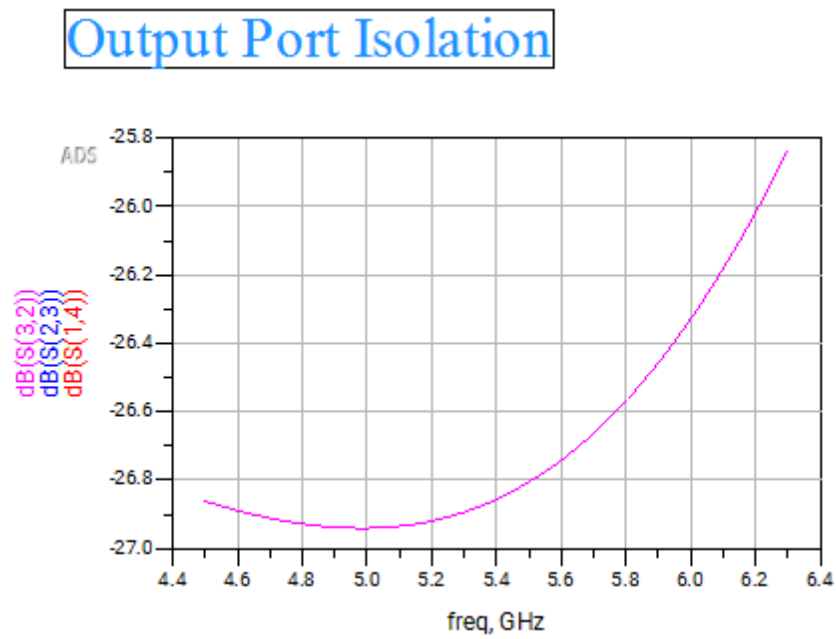


Figure 8. Lange Coupler Output Port Isolation

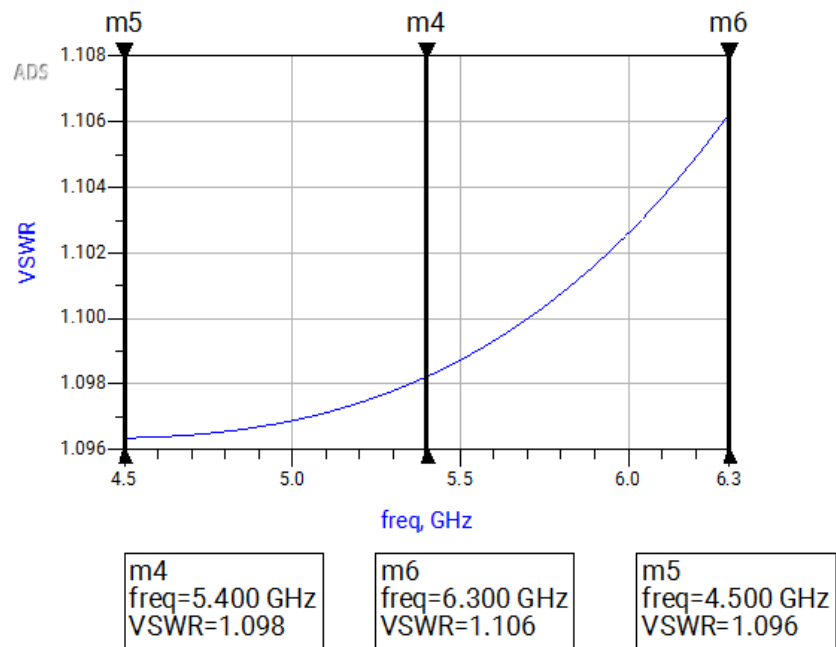


Figure 9. Lange Coupler VSWR

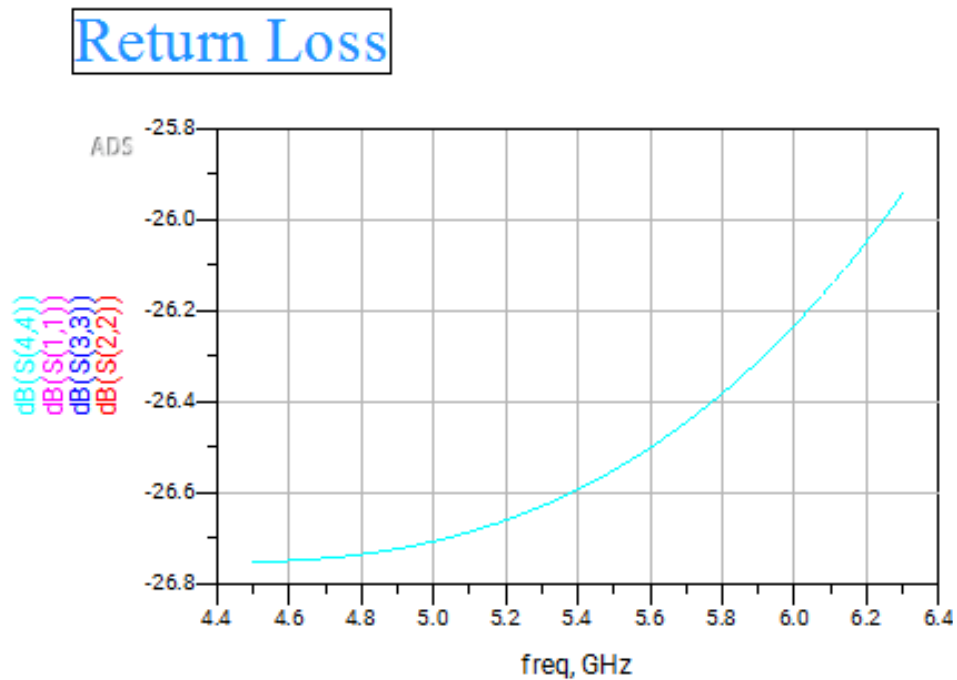


Figure 10. Lange Coupler VSWR

Insertion Loss	Phase	VSWR	Output Port Isolation	Return Loss
-3.02 dB (Port 1) -3.11 dB (Port 2)	-0.15 degree (Port 1) -90.00 degree (Port 2)	1.1:1	26.85 dB	26.6 dB

Table 1. Lange Coupler Specifications

2.2 0°/180° Phase Modulator

The second stage of the system implements binary phase modulation, producing either a 0° or 180° phase shift. A double-balanced mixer DMX0418L was selected as the phase modulation element due to its inherent port isolation, suppression of even-order harmonics, and improved linearity compared to single-ended mixers. In this configuration, the quadrature divider output connects to the mixer's LO port, while the IF port receives the binary control signal (± 1 V DC) to switch the output phase. The modulated RF output then feeds the subsequent attenuator stage. Depending on the polarity of the IF signal, the output experienced either a positive or negative sign inversion, corresponding to a 0° or 180° phase shift.

The selected mixer operates across 4-18 GHz for both RF and LO inputs, with IF capability from DC to 4 GHz. As shown in Appendix B, it exhibits low conversion loss ranging from 5.05 dB to 5.35 dB across the bandwidth (0.3 dB variation), and excellent port matching at both LO and RF ports. The mixer's LO port requires an input power range of 7–13 dBm for proper operation. Accounting for the quadrature divider's 3 dB insertion loss, the initial system input power should be approximately 10-16 dBm to ensure adequate LO drive power.

Due to the mixer's nonlinear behavior, a fully accurate simulation in ADS was not achievable. This is what also led to the phase assumption of the mixer as ADS was

incapable of showing the phase specs across our bandwidth frequency when simulating. Multiple approaches were made such as directly importing the mixer model from the Marki toolkit in ADS, but this did not work. Therefore, the mixer was modeled using an equivalent insertion loss element and an ideal phase shifter as shown in Figure 6. To account for the mixer's amplitude variation across its operating range, both minimum 5.05 dB and maximum 5.35 dB insertion loss cases were analyzed when simulating the overall circuit. An additional 1 dB insertion loss was accounted for using the LO - RF port to calculate the conversion loss to account for any device tolerance. Moreover, a $\pm 0.5^\circ$ phase error will be incorporated into the system-level analysis to account for mixer-induced phase deviations for the same reason.

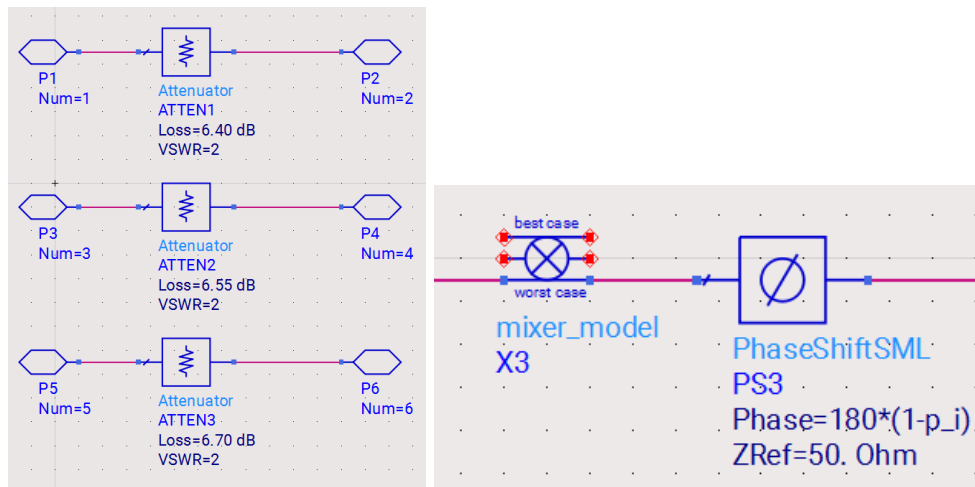


Figure 11. Mixer Implementation with an Attenuator and a PhaseShiftSML on ADS

The DMX0418L mixer features SMA ports, requiring appropriate connectors to interface with the Lange coupler output and the subsequent attenuator stage. Each connection uses an SMA connector (2921-61493, 0.13 dB insertion loss) and a 10 cm SMA cable (RG402 tinned coax, 0.04 dB insertion loss). With two connections per signal path (the Lange

coupler output to mixer LO port, and the mixer RF port to attenuator input), the total interconnection loss is approximately 0.35 dB per path. Therefore, the total conversion loss through the 0°/180° phase modulator stage ranges from 6.4 dB (minimum) to 6.7 dB (maximum), calculated as shown below:

- Minimum: 5.05 dB (mixer) + 0.35 dB (cable) + 1.0 dB (LO-to-RF path loss) = 6.4 dB
- Maximum: 5.35 dB (mixer) + 0.35 dB (cable) + 1.0 dB (LO-to-RF path loss) = 6.7 dB

2.3 Phase-Invariant Attenuator

The third stage of the modulator provides amplitude control without introducing relative phase distortion. This was achieved using a π -resistor network to produce attenuation and a matched through-line for the unattenuated path. To ensure equal spacing between constellation points, the inner constellation points were required to have an amplitude that was one-third that of the outer points. This corresponds to approximately 9.5 dB of attenuation, calculated as:

$$20\log_{10}\left(\frac{1}{3}\right) = -9.5\text{ dB}$$

PIN diodes were used to switch between the attenuated and unattenuated paths based on the control bit. Care was taken to ensure that both signal paths introduced minimal and nearly identical phase shifts. It is important to note that much tuning was done to the attenuator when testing the overall circuit, and we deduced that this is likely resulting from weak isolation of the diodes. To mitigate this, a future design implementation would have multiple diodes in parallel to boost the isolation of the circuit while minimally affecting the insertional loss.

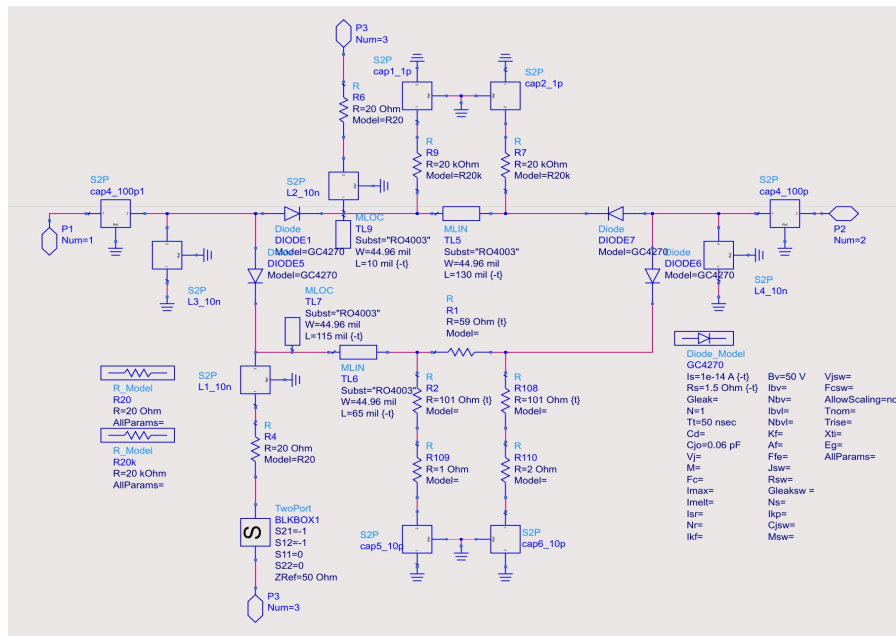


Figure 12. Phase-Invariant Attenuator Implementation

Attenuator Output Plots

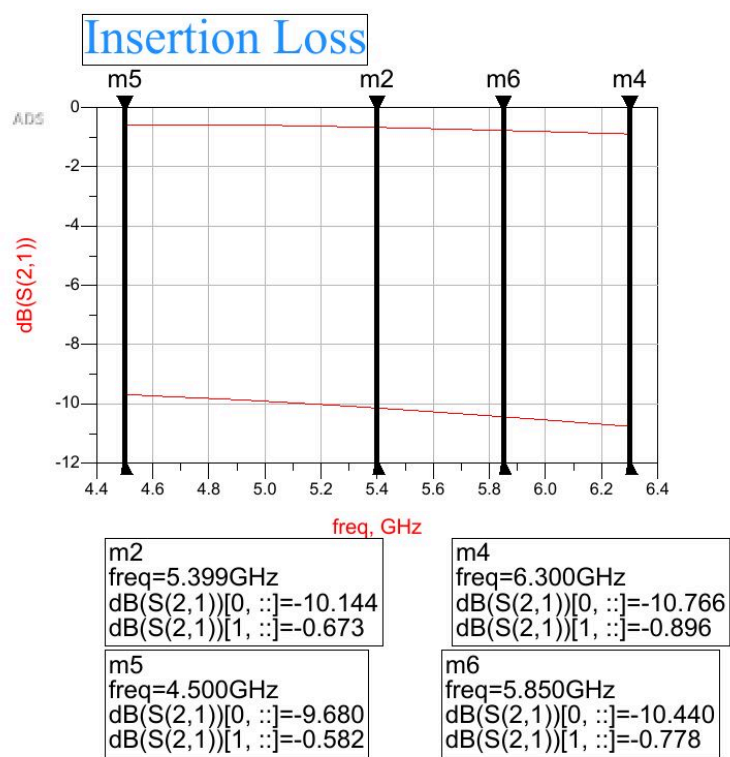


Figure 13. Attenuator Insertion Loss

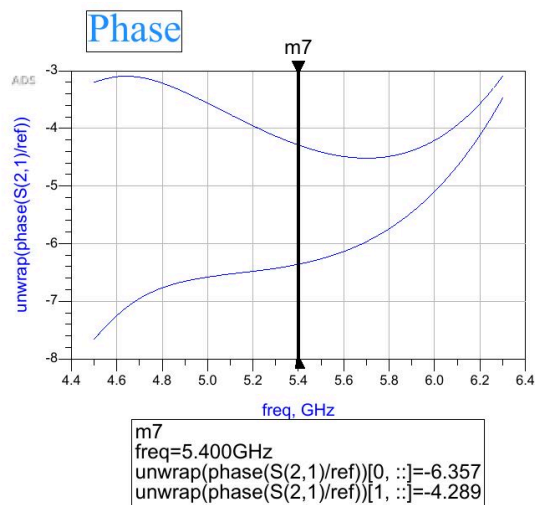


Figure 14. Attenuator Phase

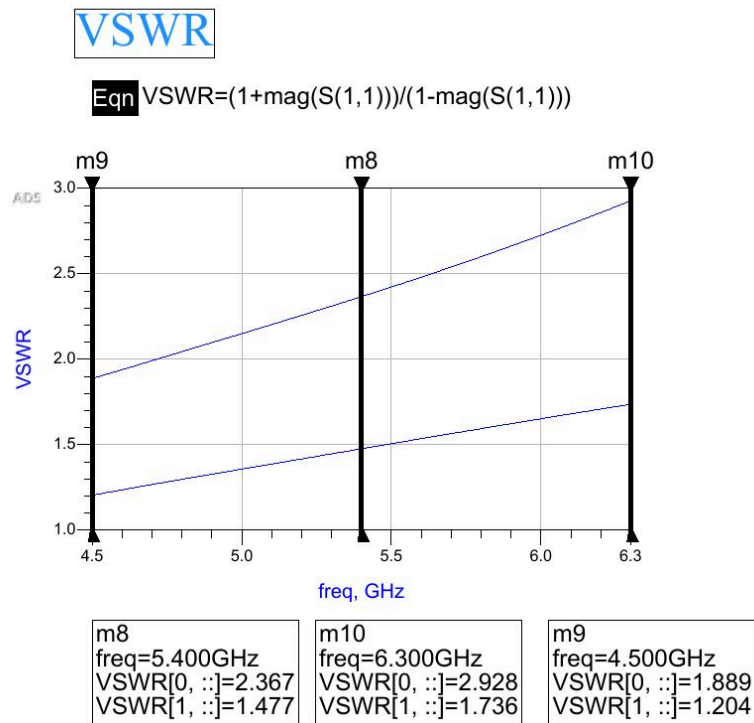


Figure 15. Attenuator VSWR



Figure 16. Attenuator Return Loss

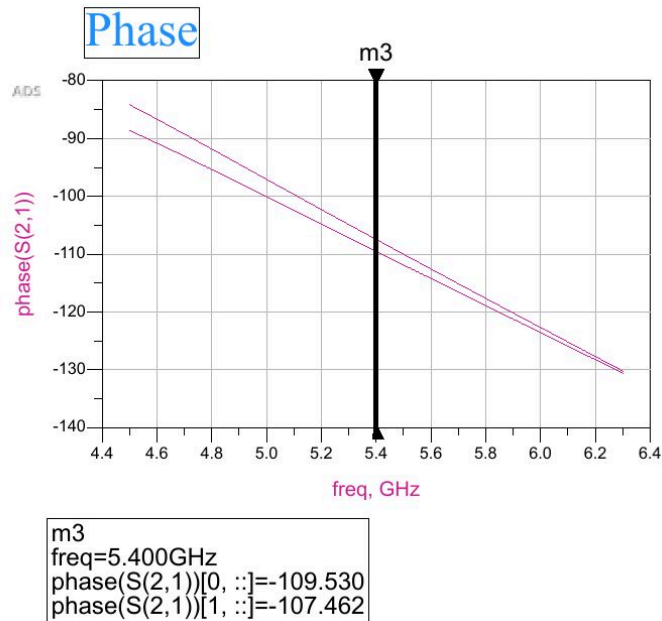


Figure 17. Attenuator Phase

2.4 In-Phase Power Combiner

The final stage of the vector modulator combines the I and Q signal paths using a Wilkinson power combiner. This topology was selected due to its excellent isolation between input ports, inherent impedance matching, and preservation of phase coherence. The combiner was carefully laid out to maintain symmetry between the two signal paths, ensuring that electrical lengths were matched and that phase imbalance was minimized.

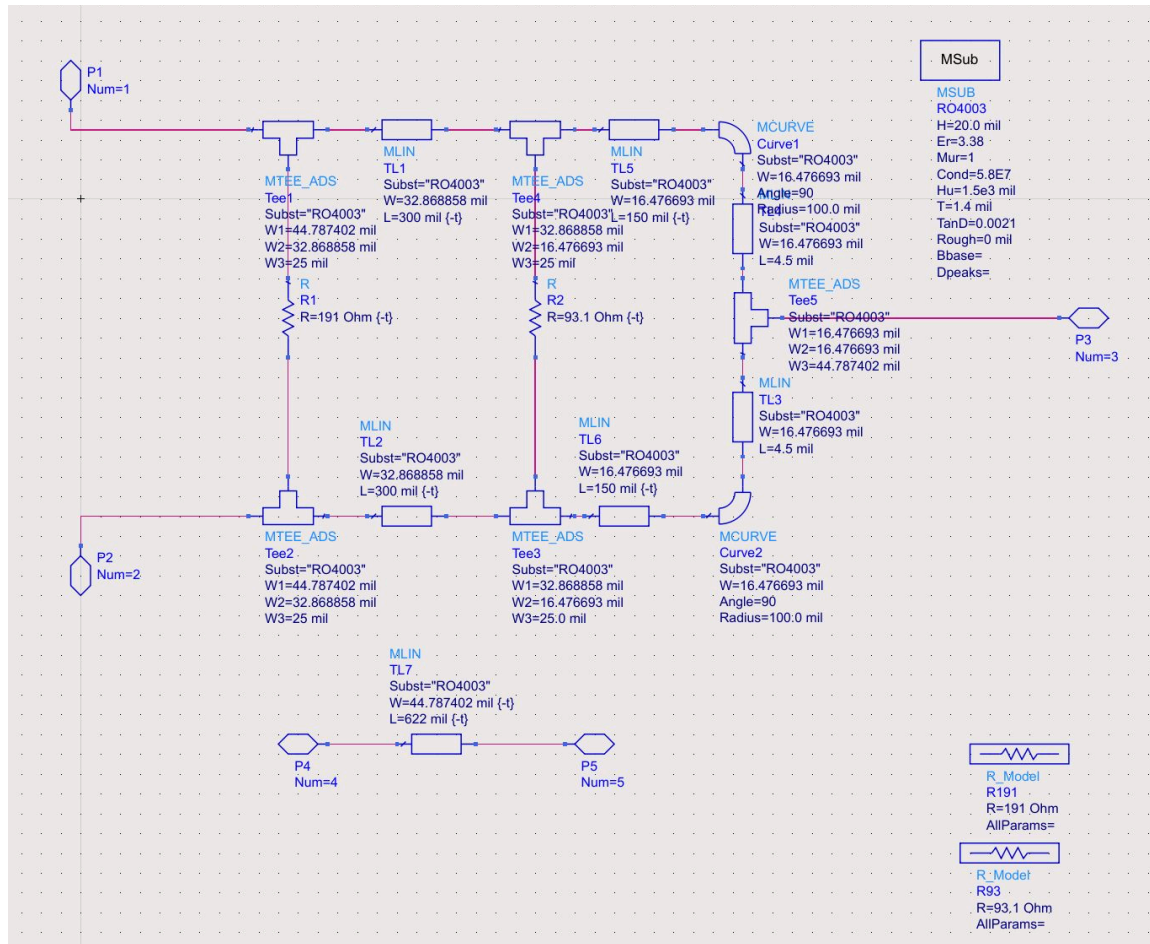


Figure 18: Wilkinson Power Combiner Schematic

Power Combiner Output Plots

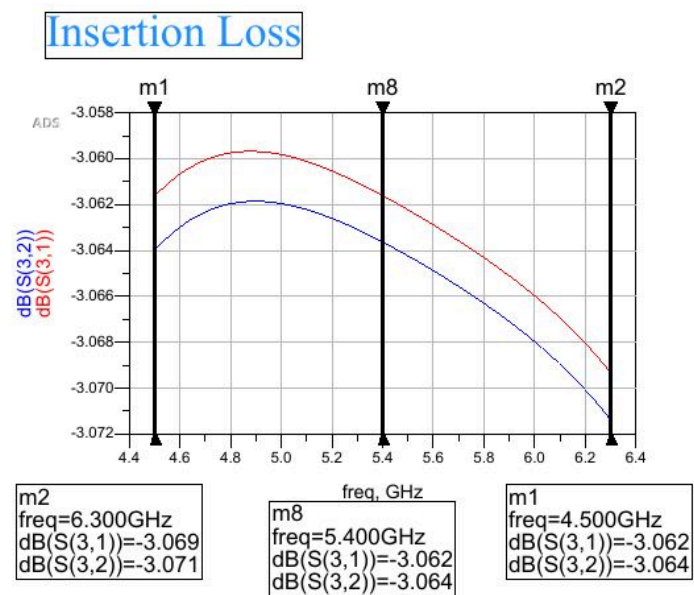


Figure 19. Combiner Insertion Loss

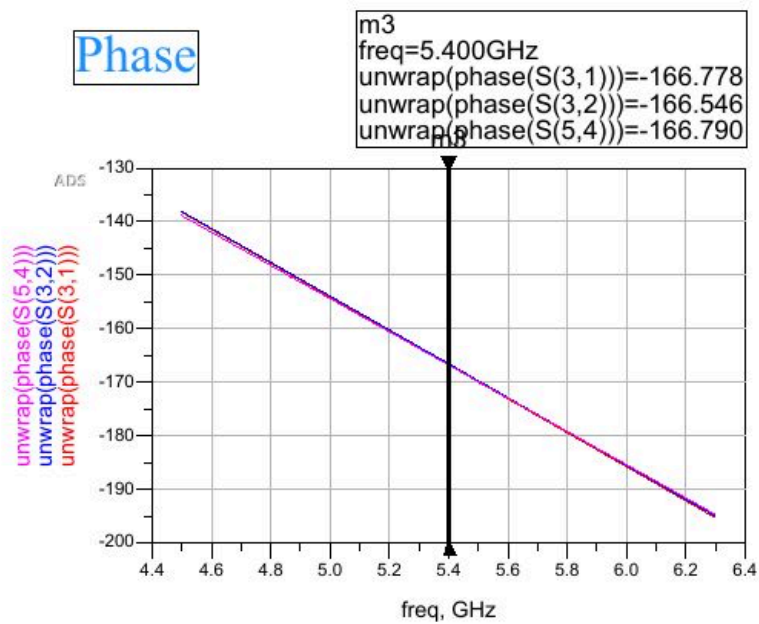


Figure 20. Combiner Phase

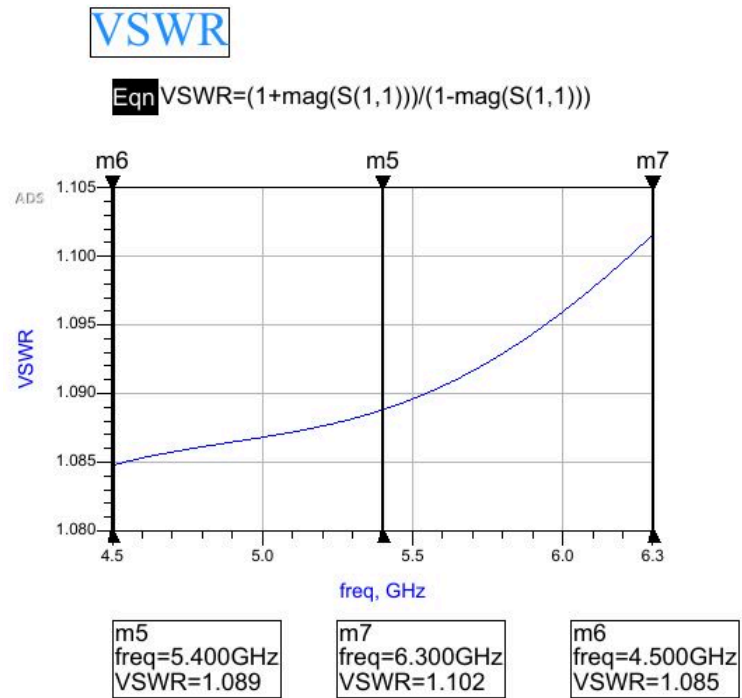


Figure 21. Combiner VSWR

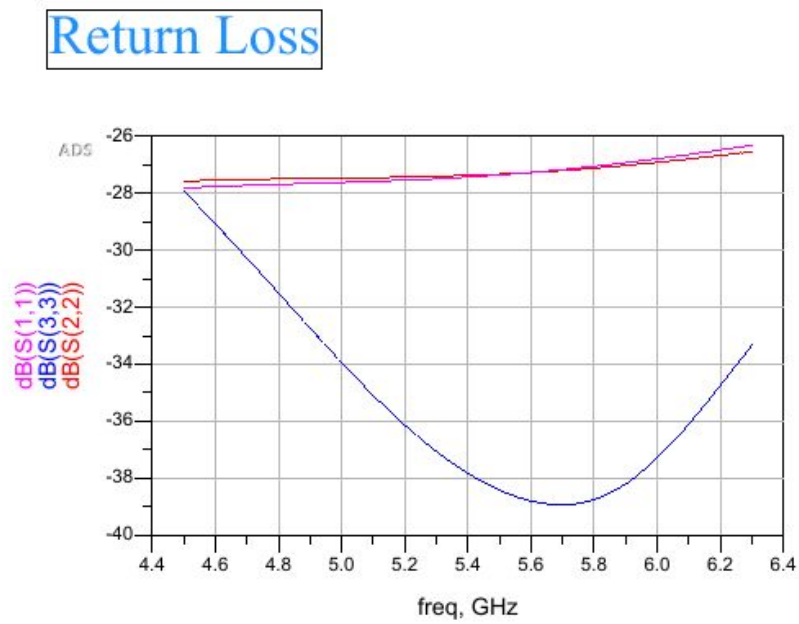


Figure 22. Combiner Return Loss

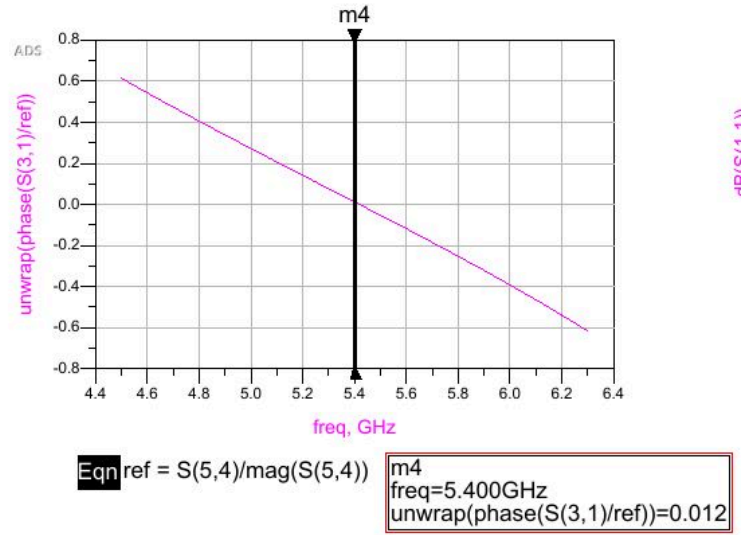


Figure 23. Combiner Unwrapped Phase

3. Results and Data Processing

The full vector modulator schematic was simulated using ADS. S-parameter data was exported for all 16 digital states and post-processed using custom MATLAB scripts. Linear phase contributions due to electrical length were removed prior to constellation analysis.

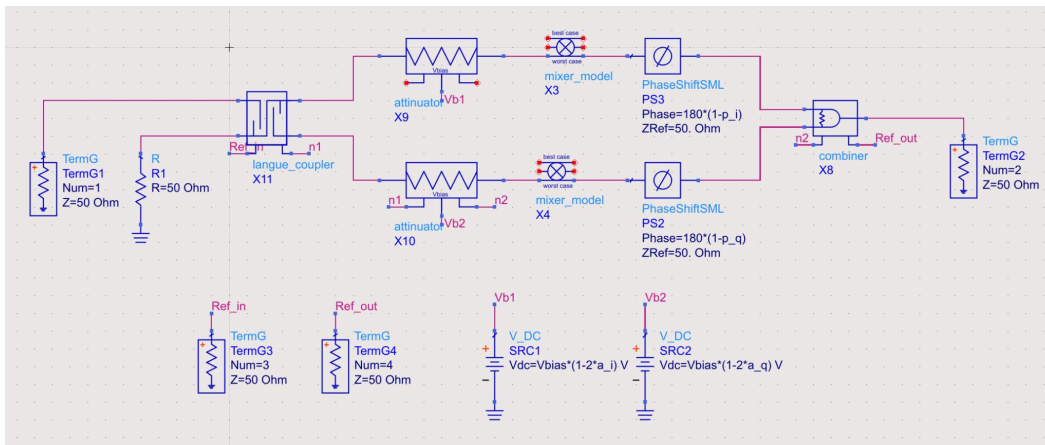


Figure 24. Top testbench

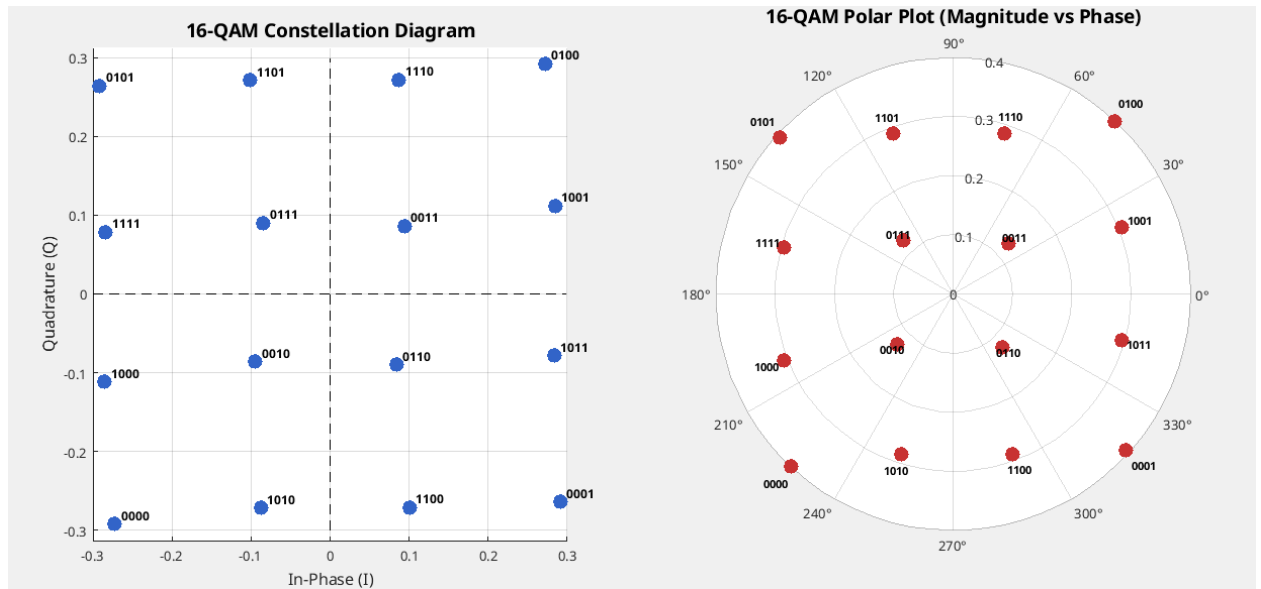


Figure 24. Constellation plots from MATLAB

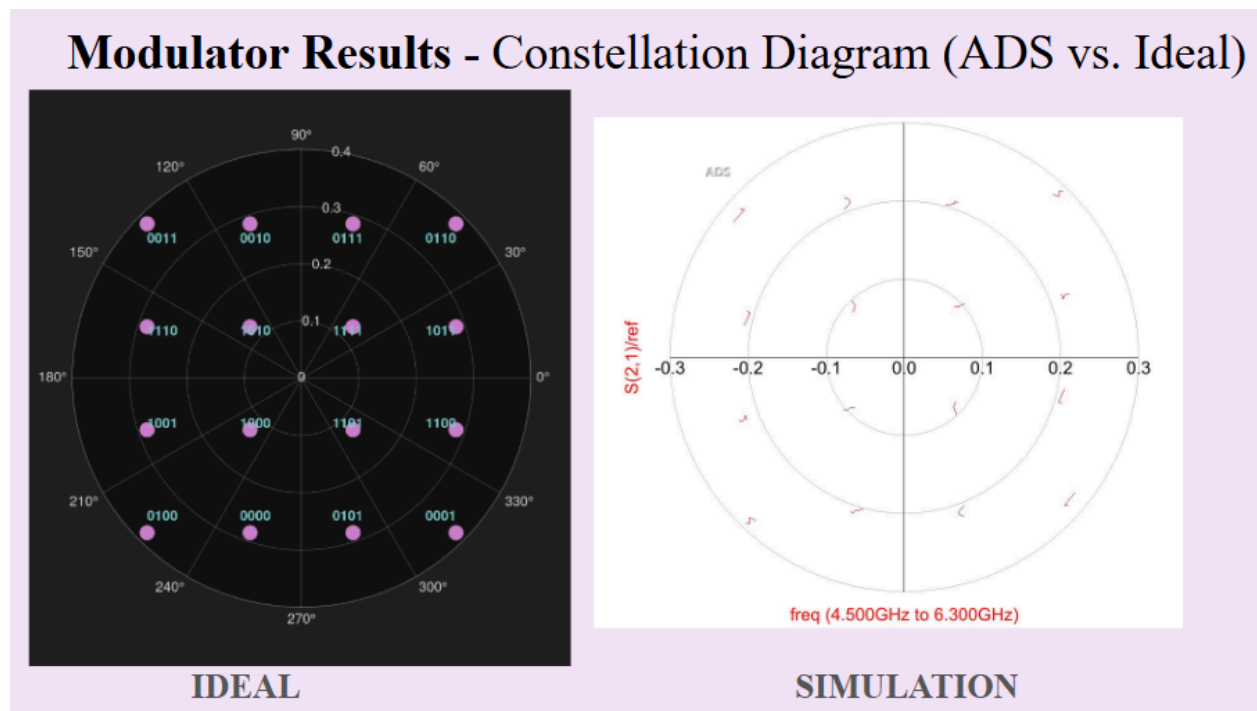


Figure 25. Constellation plots of the ideal values vs our simulation

Worst-case phase and amplitude deviations were calculated for each state relative to the ideal constellation points, and compliance with the design specifications was evaluated.

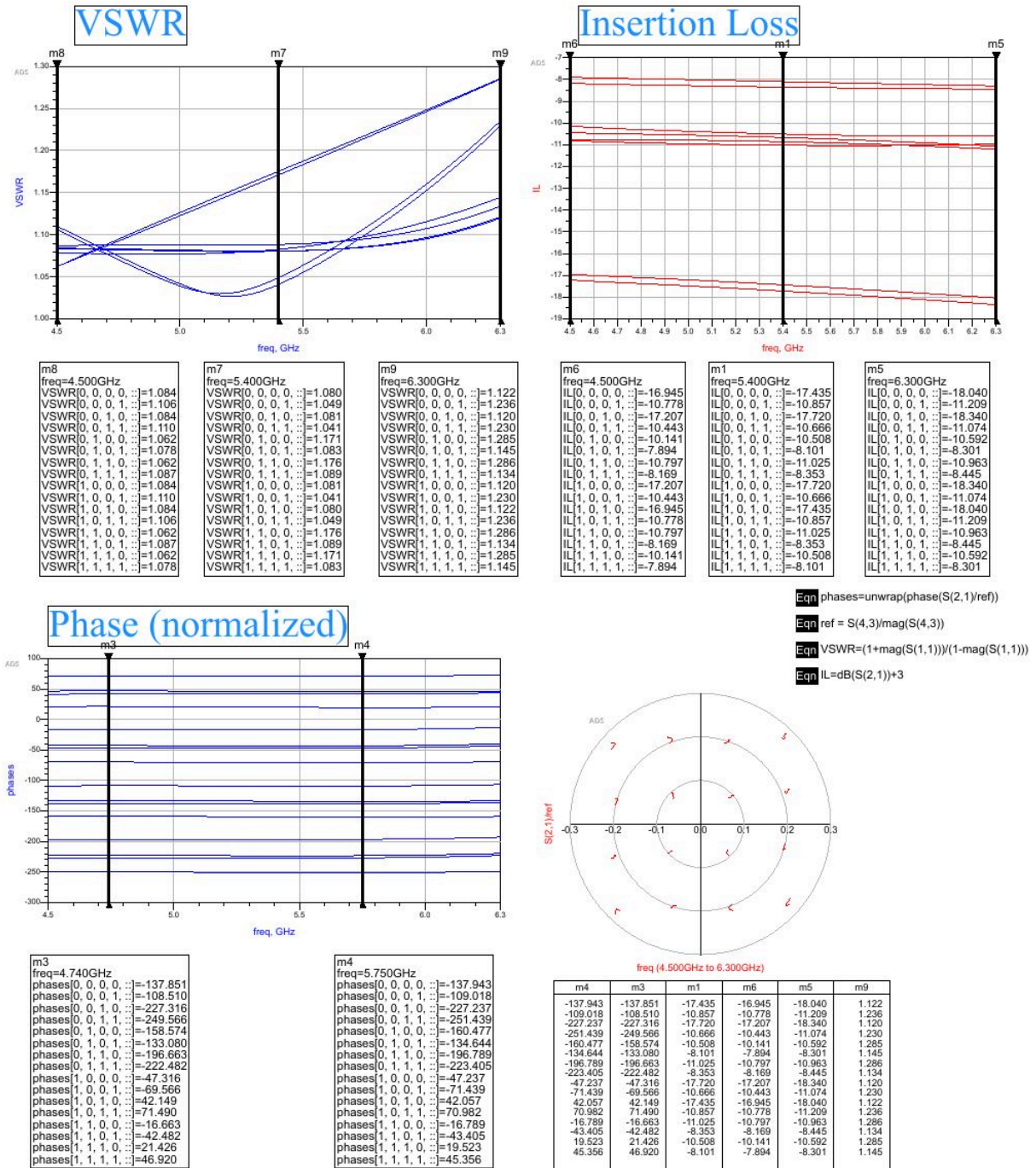


Figure 26. Overall IL, Phase, and VSRW of the top testbench

4. Performance Summary

	Achieved Specs	Required Specs
Minimum Insertion Loss	8.10 dB	$\leq 8.5\text{dB}$
Phase Accuracy	± 3.4979 degrees	$\leq \pm 3.5$ degrees
VSWR	1.33:1	$\leq 1.4:1$
Amplitude Variation vs. Frequency	$\leq .692$ dB	$\leq \pm 0.7$ dB

Symbol	Phase [deg]	Ideal Phase	Phase Error	Power [dB]	Ideal Power	Power Variation	VSWR	I	Q
1000	-47.01	-45	2.01	-18.176	-17.48992509	0.6860749056	1.125	0.08412053614	-0.09023980224
0010	132.99	135	2.01	-18.176	-17.48992509	0.6860749056	1.125	-0.08412053614	0.09023980224
0000	-137.72	-135	2.72	-16.797	-17.48992509	0.6929250944	1.127	-0.09456992743	-0.08599168866
1010	42.28	45	2.72	-16.797	-17.48992509	0.6929250944	1.127	0.09456992743	0.08599168866
1101	-42.166	-45	2.834	-8.096	-7.9475	0.1485	1.144	0.2918348892	-0.2643043252
0111	137.834	135	2.834	-8.096	-7.9475	0.1485	1.144	-0.2918348892	0.2643043252
0101	-133.03	-135	1.97	-7.651	-7.9475	0.2965	1.155	-0.2727577221	-0.2921899689
1111	46.97	45	1.97	-7.651	-7.9475	0.2965	1.155	0.2727577221	0.2921899689
1001	-69.555	-71.56505118	2.010051177	-10.777	-10.50022505	0.276774949	1.261	0.1010086649	-0.2709525656
0011	110.445	108.4349488	2.010051177	-10.777	-10.50022505	0.276774949	1.261	-0.1010086649	0.2709525656
0001	-107.889	-108.4349488	0.5459488229	-10.886	-10.50022505	0.385774949	1.267	-0.08771711488	-0.271755787
1011	72.111	71.56505118	0.5459488229	-10.886	-10.50022505	0.385774949	1.267	0.08771711488	0.271755787
1110	21.398	18.43494882	2.963051177	-9.911	-10.50022505	0.589225051	1.333	0.2854839243	0.1118684309
0100	-158.602	-161.5650512	2.963051177	-9.911	-10.50022505	0.589225051	1.333	-0.2854839243	-0.1118684309
1100	-15.437	-18.43494882	2.997948823	-10.607	-10.50022505	0.106774949	1.334	0.2842448894	-0.07849160585
0110	164.563	161.5650512	2.997948823	-10.607	-10.50022505	0.106774949	1.334	-0.2842448894	0.07849160585

Figure 27. Numerical values of the phase, power, VSWR, and necessary errors

- Maximum VSWR remained below the 1.4 requirement across all states.
- The maximum insertion loss of the outer constellation points was within the 8.5 dB limit.
- Twelve out of sixteen constellation points met both phase and amplitude accuracy requirements across the full bandwidth

5. Additional Considerations

Practical implementation considerations such as circuit size, weight, cost, and manufacturability were evaluated. While most components were inexpensive, the mixer represented the largest cost contributor. The total circuit mass and footprint were estimated based on substrate dimensions and component packaging.

6. Future Improvements

Several improvements were identified, including reducing insertion loss through improved mixer selection, refining the DC biasing network for the attenuator, and replacing ideal components with realistic lumped-element models. Additional bandwidth optimization could also be achieved by modifying the coupling characteristics of the quadrature divider. The main additional consideration, however, was improving the isolation of the attenuator through the use of parallel diodes. Due to this weakened isolation, the attenuator was having signal leakage that affected the overall circuit, thus leading to tedious tuning of the attenuator.

7. Conclusion

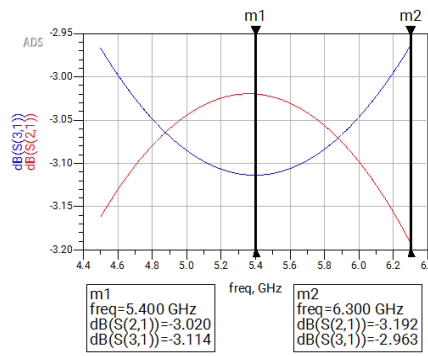
Although not all constellation points met the specified accuracy requirements across the full bandwidth, the vector modulator successfully demonstrated the feasibility of a fully passive 16-QAM implementation at 5.4 GHz. The project highlighted the importance of system-level interactions between cascaded RF blocks and emphasized the trade-offs inherent in broadband passive designs. The project made us appreciate the reality of system design, not just homework

problems, but choosing real components that face nonidealities that interact with each other to cause loss and phase divergence.

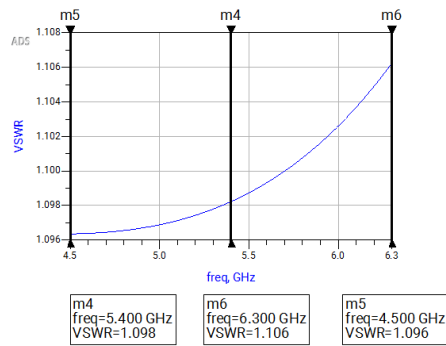
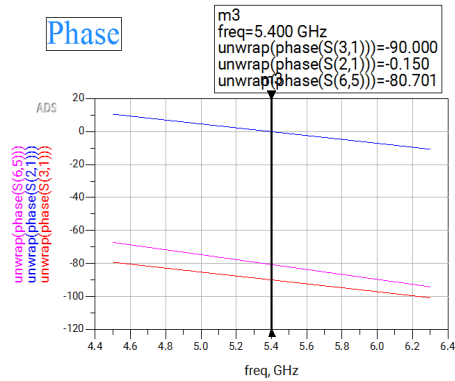
8. Appendix

Appendix A. Insertion Loss, Phase, VSWR, Isolation, and Return Loss of Lange Coupler

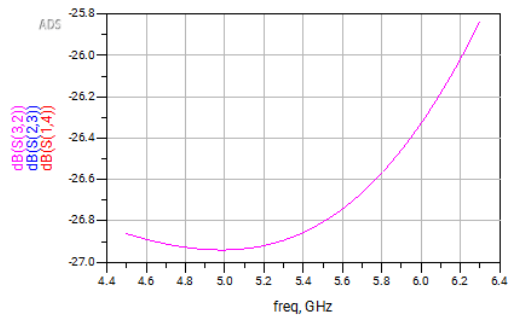
Insertion Loss



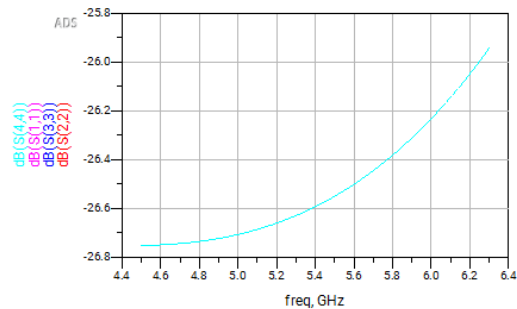
Phase



Output Port Isolation



Return Loss



Appendix B. DMX0418L Datasheet

4 TO 18 GHz DOUBLE-BALANCED MIXER

MODELS: DMX0418L, DMY0418L, DMX0716L AND DMY0716L

FEATURES

- RF/LO coverage 4 to 18 GHz
- IF operation DC to 4 GHz
- LO power range +7 to +13 dBm
- Conversion loss 5 dB typical
- LO-to-RF Isolation 35 dB typical
- Packaging Hermetically sealed



MITEQ's DMX and DMY Series of mixers are constructed using double-tuned microstrip RF and LO baluns with a DC-coupled IF structure. The construction, coupled with the hermetic packaging, provides for high inherent reliability and isolation over an extremely broad frequency range. This device performs as an up- or downconverter covering most EW bands and communication applications. This mixer is also available with medium or high forward voltage diodes (M, H) yielding proportional changes in LO power and spurious performance.

ELECTRICAL SPECIFICATIONS

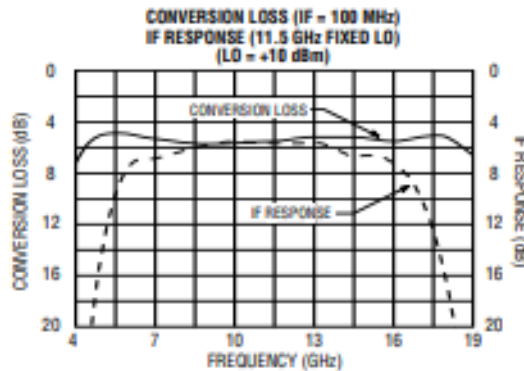
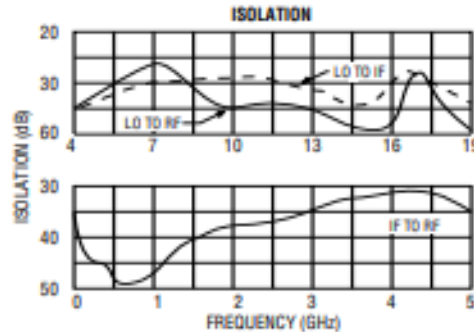
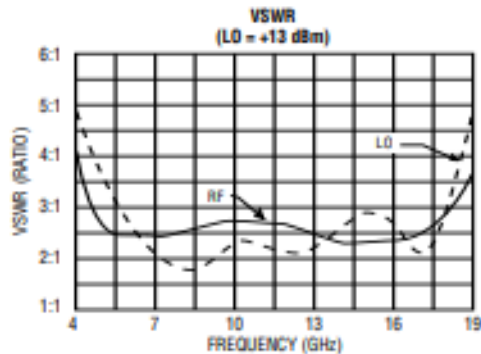
INPUT PARAMETERS	CONDITION	UNITS	MIN.	TYP.	MAX.
RF frequency range		GHz	4		18
RF VSWR (RF = -10 dBm, LO = +10 dBm)	7 to 16 GHz 4 to 18 GHz	Ratio Ratio		2:1 3:1	
LO frequency range		GHz	4		18
LO power range		dBm	+7		+13
LO VSWR (LO = +10 dBm)	7 to 16 GHz 4 to 18 GHz	Ratio Ratio		2:1 2.5:1	
TRANSFER CHARACTERISTICS	CONDITION	UNITS	MIN.	TYP.	MAX.
Conversion loss (IF = 100 MHz, LO = +10 dBm)	7 to 16 GHz 4 to 18 GHz	dB dB		5 6	8 9
Single-sideband noise figure	4 to 18 GHz	dB			9.5
LO-to-RF isolation	4 to 18 GHz	dB	20	35	
LO-to-IF isolation	4 to 18 GHz	dB	20	35	
IF-to-RF isolation	DC to 4 GHz	dB	20	35	
Input power at 1 dB compression	LO = +10 dBm	dBm	0	+3	
Input two-tone third-order intercept point	LO = +13 dBm	dBm	+10	+13	
OUTPUT PARAMETERS	CONDITION	UNITS	MIN.	TYP.	MAX.
IF frequency range	2 dB bandwidth	GHz	DC		4
IF VSWR (IF = -10 dBm, LO = +10 dBm)		Ratio		2:1	



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DMX0418L/DMY0418L TYPICAL TEST DATA



SINGLE-TONE (m) RF x (n) LO RELATIVE SPUR LEVEL (dBc)
(AVERAGE MIDBAND RF, LO, IF FREQUENCIES,
RF = -10 dBm, LO = +10 dBm)

SPUR (m) RF x (n) LO	RF TEST FREQ. (GHz)	LO TEST FREQ. (GHz)	SPUR LEVEL (dBc)
1 x 1	10	12	0
1 x 2	14	8	32
1 x 3	16	6	12
2 x 1	6	14	38
2 x 2	10.5	11.5	55
2 x 3	12.8	9.2	38
3 x 1	5	17	42
3 x 2	8.4	13.6	60
3 x 3	10.7	11.3	58

MAXIMUM RATINGS

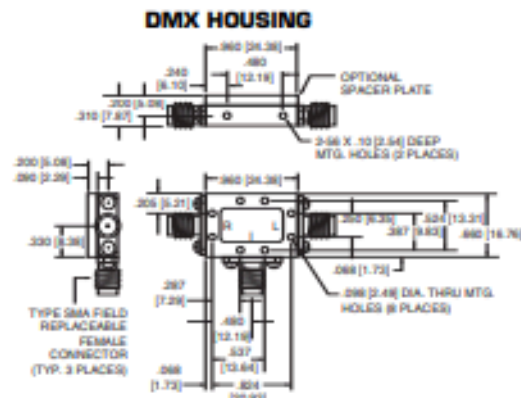
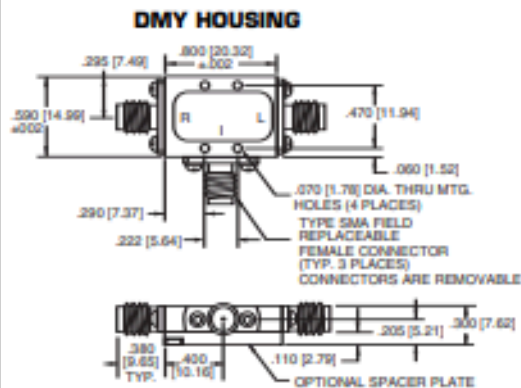
Specification temperature +25°C
Operating temperature -54 to +85°C
Storage temperature -65 to +125°C

AVAILABLE OPTIONS

Medium/high dynamic range options
M (LO = +13 to +17 dBm), (IP₃ = +18 dBm typ.)
H (LO = +17 to +20 dBm), (IP₃ = +22 dBm typ.)
DMX0418, M, H (Conversion loss = 10 dB max.)
DMX0716, M, H (Conversion loss = 9 dB max.)

NOTE: Test data supplied at 25°C; conversion loss and LO-to-RF isolation.

OUTLINE DRAWINGS



NOTE: All dimensions shown in brackets [] are in millimeters.



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